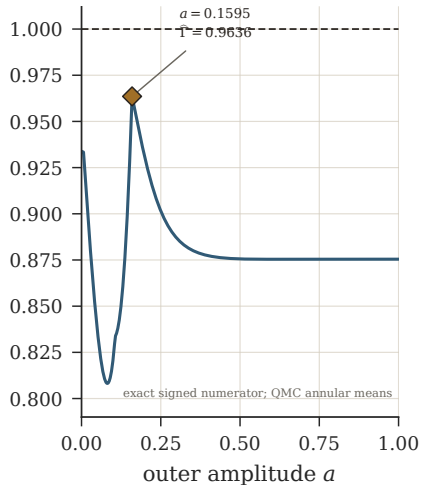
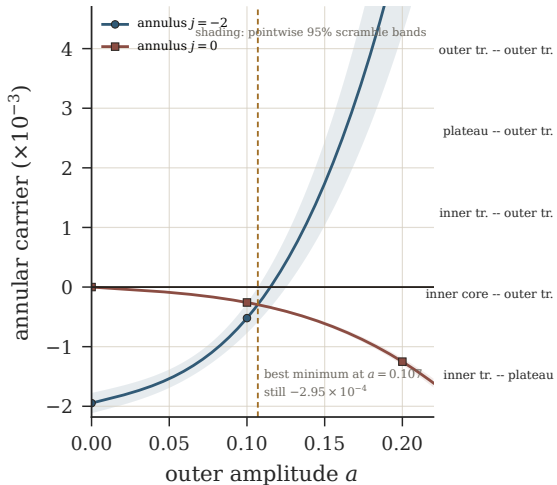


a Full-annulus screen



b Coarse-annulus root gap



c Negative $j = 0$ mechanism

